Kaleem Nawaz Khan

kk5271@rit.edu | LinkedIn |G. Scholar | Website

Areas of Interest: Networked Sensing, Autonomous Systems, Cooperative Perception

---

Research Focus

My research focuses on collaborative perception between vehicles and roadside infrastructure. It enables sensing beyond line-of-sight. I study the co-design of perception, communication, and computation. The goal is reliable and safe autonomous decision-making. My work has appeared in top venues such as MobiSys and SenSys.

Education

Rochester Institute of Technology (RIT)*, Rochester, New York*
PhD Computing and Information Sciences — *Aug. 2022 – Present*

National University of Computer and Emerging Sciences*, Peshawar, KPK*
Master of Science (Computer Science) — *Aug. 2015 – Dec. 2017*

Research Experience

Research & Development Center, General Motors*, Warren, Michigan*
Research Intern (Mentor: Paolo Giusto) — *May. 2024 – Aug. 2024*

- *Developed solutions to enable smart off-loading of vehicular data to edge/cloud*
- *Explored OMNET++ with Simu5G integration to investigate different network scenarios for vehicle-to-edge communication and data collection*

Networked Sensing Systems Lab, RIT*, Rochester, New York*
Graduate Research Assistant (Advisor: Fawad Ahmad) — *Aug. 2022 – Present*

- *I develop fast, accurate, and scalable infrastructure-assisted and vehicle-to-vehicle cooperative perception systems.*
- *Additionally, I also explore collaborative, view-based 3D scene representations to enhance vehicles' perception beyond traditional approaches.*

National Center of Artificial Intelligence*, Peshawar, KPK*
Research Team Lead (PI: M. Salman Khan) — *Apr. 2019 – Sep. 2020*

- *Led a diverse team in R&D to develop AI solutions for healthcare.*
- *We utilized deep learning and related technologies for screening to assist cardiology and hematology practitioners.*

Research Projects

FRC: Radar–Camera Fusion for Intersection Awareness
Radar, Camera, VLMs, Multimodal Sensor Fusion — *Nov. 2025 – Present*

- *Exploring solutions to map sparse radar data onto 2D images and fusing complementary information reliably.*
- *Accurate Spatial Alignment and time synchronization to correctly match radar detections with camera frames.*

PMC: Collaborative Intersection Perception Using Multi-Vehicle Cameras
SfM, 3D Reconstruction, Spatial Reasoning, Segmentation — *May. 2025 – Present*

- *Creating and real-time updating SfM-based 3D models of the intersection using images from vehicle-mounted cameras.*
- *Estimating dynamic objects in the scene based on object segmentation.*
- *Performing spatial reasoning on the SfM-based 3D model to identify blind spots and determine the most relevant information for vehicles to exchange.*

ARC: Accurate, Real-time, and Scalable Multi-vehicle Cooperative Perception
C++, LiDAR, PCL, CUDA, CarLA — *Jun. 2024 – Mar. 2025*

- *Enabled high-accuracy cooperative perception while preserving scalability, addressing a key gap in existing systems.*
- *Added a refinement layer for vehicle point cloud alignment, reducing errors in existing methods.*
- *Employed grid-based spatial reasoning to minimize alignment latency and enable selective data sharing among vehicles.*

Smart Vehicle-to-Edge Computation and Data Offloading
OMNET++, Simu5G, MEC, ML — *May. 2024 – Aug. 2024*

- *Enabled smart off-loading of vehicular data and apps to edge/cloud*
- *Used Simu5G to simulate complex 5G network scenarios involving moving vehicles, base stations, and edge servers.*

- *Generated simulated data on network latency, signal strength, and congestion under various conditions.*

VRF: Vehicle Road-side Point Cloud Fusion

ROS, FAST-LIO, PCL, Open3D, Docker Sep. 2022 – Mar. 2024

- *Tackle vehicles' sensor occlusion by enabling real-time sharing and accurate fusion of raw 3D data from roadside LiDARs.*
- *Decoupled vehicle and roadside point cloud alignment by aligning both to a 3D map and then enabling a direct alignment.*
- *Evaluation on real-world testbeds demonstrates VRF can fuse vehicle roadside point clouds with 20ms end-to-end latency and 5 cm positioning accuracy, thereby multiplying the vehicle's reaction time by a factor of 5.*

Publications

Conferences

- ***Khan, K. N.***, *& Ahmad, F. (2025). ARC: Accurate, Real-time, and Scalable Multi-vehicle Cooperative Perception. In Proceedings of the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (**SenSys 2026**).*
- ***Khan, K. N.***, *Khalid, A., Turkar, Y., Dantu, K., & Ahmad, F. (2024, June). VRF: Vehicle Road-side Point Cloud Fusion. In Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (**MobiSys 2024**).*
- *Hasib, R.,* ***Khan, K. N.***, *Yu, M., & Khan, M. S. (2021, April). Vision-based human posture classification and fall detection using convolutional neural networks. In 2021 International Conference on Artificial Intelligence (ICAI) (pp. 74-79). IEEE.*
- *Qasim, S.,* ***Khan, K.N.***, *Yu, M. and Khan, M.S., 2021, April. Performance evaluation of background subtraction techniques for video frames. In 2021 International Conference on Artificial Intelligence (ICAI) (pp. 102-107). IEEE.*
- *Ahmad, B., Khan, F.A.,* ***Khan, K.N.*** *and Khan, M.S., 2021, December. Automatic classification of heart sounds using long short-term memory. In 2021 15th International Conference on Open Source Systems and Technologies (ICOSST) (pp. 1-6). IEEE.*
- *Islam, M.,* ***Khan, K.N.*** *and Khan, M.S., 2021, April. Evaluation of preprocessing techniques for U-Net based automated liver segmentation. In 2021 International Conference on Artificial Intelligence (ICAI) (pp. 187-192). IEEE.*

Journals

- ***Khan, K.N***, *Khan, F.A., Abid, A., Olmez, T., Dokur, Z., Khandakar, A., Chowdhury, M.E. and Khan, M.S., 2021. Deep learning based classification of unsegmented phonocardiogram spectrograms leveraging transfer learning. Physiological measurement, 42(9), p.095003.*
- ***Khan, K.N.***, *Ullah, N., Ali, S., Khan, M.S., Nauman, M. and Ghani, A., 2022. Op2Vec: An Opcode Embedding Technique and Dataset Design for End-to-End Detection of Android Malware. Security and Communication Networks, 2022(1), p.3710968.*
- *Salman Khan, M., Ullah, A.,* ***Khan, K.N.***, *Riaz, H., Yousafzai, Y.M., Rahman, T., Chowdhury, M.E. and Abul Kashem, S.B., 2022. Deep Learning Assisted Automated Assessment of Thalassaemia from Haemoglobin Electrophoresis Images. Diagnostics, 12(10), p.2405.*
- *Zeng, A., Wu, C., Lin, G., Xie, W., Hong, J., Huang, M., Zhuang, J., Bi, S., Pan, D., Ullah, N. and* ***Khan, K.N***, *2023. Imagecas: A large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images. Computerized Medical Imaging and Graphics, 109, p.102287.*

Selected Talks

VRF: Vehicle Road-side Point Cloud Fusion, *Tokyo, Japan*

ACM MobiSys 2024, International Conference on Mobile Systems, Applications, and Services 3 – 7 June. 2024